

AI-Powered Enterprise E-Commerce Platform

Stakeholder Analysis Document

1. Introduction

Before developing any software system, it is important to identify the people, groups, and systems that are connected to the project.

These people or groups are called **stakeholders**.

A stakeholder is anyone who:

- uses the system,
- manages the system,
- develops the system,
- benefits from the system,
- or is affected by the system.

Understanding stakeholders helps developers:

- understand requirements clearly,
- reduce misunderstandings,
- build useful features,
- improve customer satisfaction,
- and reduce project risks.

2. Definition of Stakeholders

A stakeholder is:

"Any individual, group, or organization that can affect or be affected by the achievement of a project's objectives."

Stakeholders can be divided into two main categories:

1. Internal Stakeholders

People directly involved in building and operating the system.

Examples:

- Developers
- Administrators

- Business Owners
 - System Analysts
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2. External Stakeholders

People outside the development team who use or are affected by the system.

Examples:

- Customers
 - Suppliers
 - Delivery Companies
 - Payment Providers
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3. Importance of Stakeholder Analysis

Stakeholder analysis helps to:

Understand user needs.

Identify project risks.

Improve communication.

Prioritize system requirements.

Increase project success rate.

Without proper stakeholder analysis, developers may build features that users do not need.

4. Stakeholders of the Proposed System

The following stakeholders have been identified for the AI-Powered Enterprise E-Commerce Platform.

5. Primary Stakeholders

Primary stakeholders directly interact with the system.

5.1 Customers

Description

Customers are individuals who use the platform to purchase products.

They are the most important stakeholders because the success of the platform depends heavily on customer satisfaction.

Responsibilities

Customers can:

- Register accounts
 - Login to the system
 - Search products
 - Add products to cart
 - Place orders
 - Write reviews
 - View recommendations
 - Communicate with AI assistant
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Needs and Expectations

Customers expect:

- Easy navigation
 - Fast response times
 - Secure payments
 - Accurate recommendations
 - Personalized shopping experience
 - Reliable order tracking
-

Problems Faced

Customers may face:

- Difficulty finding products
 - Irrelevant recommendations
 - Slow websites
 - Fraud concerns
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Benefits from the System

- Better recommendations
 - Faster shopping experience
 - Personalized services
 - Improved satisfaction
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5.2 System Administrator

Description

Administrators manage the entire platform.

They ensure that all modules operate properly.

Responsibilities

Administrators can:

- Manage users
 - Manage products
 - Manage categories
 - Monitor transactions
 - Generate reports
 - Monitor AI services
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Needs and Expectations

Administrators expect:

- Easy management tools
 - Security controls
 - Analytical dashboards
 - Fraud alerts
 - Inventory monitoring
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Benefits from the System

- Easier system management
 - Better business insights
 - Reduced operational effort
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5.3 Business Owner

Description

Business owners are responsible for the success and profitability of the company.

Responsibilities

- Strategic decision making

- Sales monitoring
 - Revenue analysis
 - Resource allocation
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Expectations

Business owners expect:

- Increased sales
 - Reduced losses
 - Accurate forecasting
 - Customer behavior insights
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Benefits

- Better decision making
 - Increased profits
 - Reduced risks
-

5.4 Sellers / Inventory Managers

Description

These stakeholders manage products and inventory.

Responsibilities

- Product updates
 - Price management
 - Inventory management
 - Stock monitoring
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Expectations

- Demand forecasting
 - Inventory analytics
 - Product performance reports
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Benefits

- Better stock management
 - Reduced inventory costs
 - Reduced product shortages
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6. Secondary Stakeholders

Secondary stakeholders do not use the system continuously but are affected by it.

6.1 Delivery Personnel

Description

Responsible for delivering products to customers.

Responsibilities

- Receive delivery requests
 - Update delivery status
 - Confirm deliveries
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Expectations

- Accurate order information
 - Delivery tracking system
 - Updated customer information
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Benefits

- Easier delivery management
 - Reduced delivery errors
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6.2 Payment Service Providers

Description

Organizations that process online payments.

Examples include:

- PayPal
 - Stripe
 - Visa
 - MasterCard
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Responsibilities

- Process transactions
 - Verify payments
 - Handle refunds
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Expectations

- Secure integration
 - Reliable transaction processing
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Benefits

- Increased transaction volume
 - Secure business relationships
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6.3 Suppliers

Description

Suppliers provide products to the business.

Expectations

- Inventory information
 - Product demand information
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Benefits

- Better supply planning
 - Improved coordination
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7. Technical Stakeholders

These stakeholders are responsible for developing and maintaining the system.

7.1 Software Developers

Responsibilities

- Build backend systems
- Build frontend systems
- Develop APIs

- Fix bugs
 - Maintain software
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Expectations

- Clear requirements
 - Proper documentation
 - Scalable architecture
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Benefits

- Easier development process
 - Better maintainability
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7.2 Data Scientists / AI Engineers

Responsibilities

- Build recommendation systems
 - Develop fraud detection models
 - Create forecasting models
 - Develop customer segmentation systems
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Expectations

- Clean datasets
 - Proper data pipelines
 - Sufficient computing resources
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Benefits

- Real-world AI implementation
 - Better model performance
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7.3 Database Administrators

Responsibilities

- Database design
- Performance optimization
- Backup and recovery
- Security management

Expectations

- Well-designed schemas
 - Efficient queries
 - Data consistency
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8. AI Stakeholders

Although AI modules are not humans, they are important components that influence the system.

8.1 Recommendation Engine

Purpose:

Recommend products to customers based on behavior and preferences.

8.2 Fraud Detection System

Purpose:

Detect suspicious transactions and reduce financial losses.

8.3 Forecasting System

Purpose:

Predict future product demand and sales trends.

8.4 Customer Segmentation System

Purpose:

Group customers according to purchasing behavior.

8.5 AI Chat Assistant

Purpose:

Provide intelligent customer support.

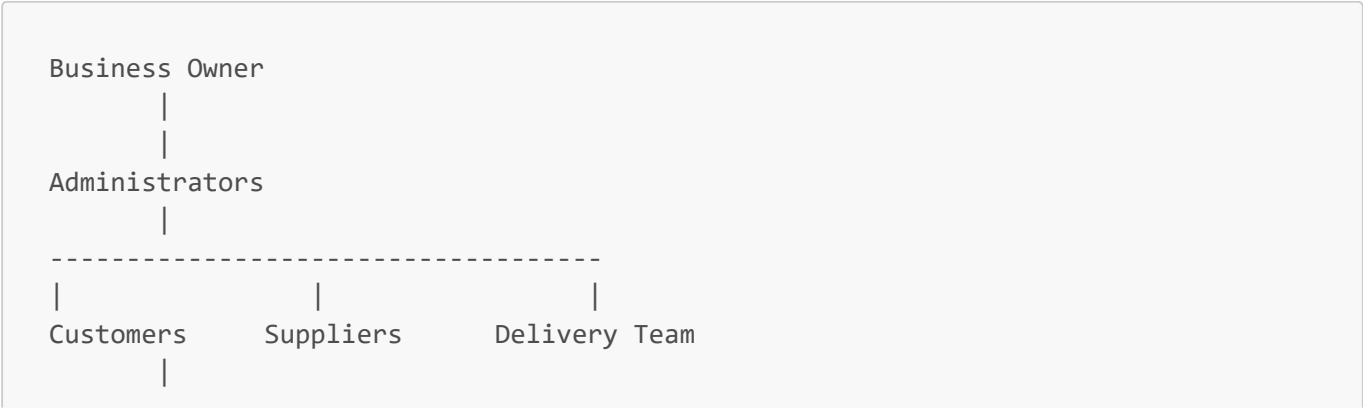
9. Stakeholder Influence and Interest Analysis

Stakeholder	Influence	Interest
Customers	High	High
Business Owner	Very High	Very High
Administrators	High	High
Developers	Medium	High
AI Engineers	Medium	High
Delivery Personnel	Medium	Medium
Suppliers	Medium	Medium
Payment Providers	High	Medium

10. Stakeholder Communication Requirements

Stakeholder	Information Needed
Customers	Order updates, recommendations
Admin	Reports, alerts
Business Owner	Sales analytics, forecasts
Developers	Requirements and technical documentation
AI Engineers	Data and model metrics
Suppliers	Demand information
Delivery Personnel	Shipping details

11. Stakeholder Diagram (Conceptual)



12. Conclusion

Stakeholder analysis is one of the most important activities in software engineering because it helps identify everyone who influences or is influenced by the system.

For the proposed AI-Powered Enterprise E-Commerce Platform, stakeholders include:

- Customers
- Business Owners
- Administrators
- Developers
- AI Engineers
- Suppliers
- Delivery Personnel
- Payment Providers

Understanding these stakeholders helps ensure that the system satisfies business needs, improves customer experience, and supports future growth.